

Name: _____

This practice exam covers Chapters 7, 8 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Identify the hybridization of the central atom in SF_6 .

ANSWER: _____

2. Which bond is longer: the N–N in N_2H_4 or the $\text{N}\equiv\text{N}$ in N_2 ? Explain.

ANSWER: _____

3. Identify the hybridization of Xe in XeF_4 and state the molecular geometry.

ANSWER: _____

4. State the type(s) of bonding present in N_2 . Explain your answer.

ANSWER: _____

5. Classify silicon carbide (SiC) as a molecular (covalent) solid, a covalent network solid, an ionic solid, or a metallic solid, and predict whether it has a relatively high or low melting point. Explain.

ANSWER: _____

6. Why is a π bond generally weaker than a σ bond?

ANSWER: _____

7. Do d orbitals participate in sp^3 hybridization? If so, how many? Justify your answer.

ANSWER: _____

8. Without calculating exact energies, which hybrid lies closer in energy to the pure $2p$ orbitals, an sp hybrid or an sp^2 hybrid? (The $2s$ orbital is lower in energy than the $2p$ orbitals.) Explain.

ANSWER: _____

9. Does the central atom in SiH_4 obey the octet rule? If not, classify the exception as electron-deficient or expanded octet, and state the number of electrons around the central atom.

ANSWER: _____

10. Draw a valid Lewis structure for SiH_4 . You must show all existing lone pairs and lone electrons.

ANSWER: _____

11. A student claims that two p orbitals always form a π bond when they overlap. Evaluate this claim by explaining what actually determines whether the bond is σ or π . Explain your reasoning completely. Answers without a correct explanation will not receive full credit.

ANSWER: _____

12. State the type(s) of bonding present in NaCl . Explain your answer.

ANSWER: _____

13. According to valence bond (VB) theory, what physically happens when a covalent bond forms between two atoms?

ANSWER: _____

14. For the reaction $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$, use average bond energies to determine the total energy required to break all of the bonds in the reactants. Average bond energies (kJ/mol): $\text{H}-\text{H} = 436$, $\text{H}-\text{I} = 298$, $\text{I}-\text{I} = 151$.

ANSWER: _____

15. Determine the hybridization of Be in BeCl_2 . State the molecular geometry and bond angle.

ANSWER: _____

16. True or false: Hybridization is a physical process that an atom carries out before it bonds. Explain.

ANSWER: _____

17. Do d orbitals participate in sp hybridization? If so, how many? Justify your answer.

ANSWER: _____

18. The species HCO_2^- exhibits resonance. Draw all of its equivalent resonance structures. What is the bond order of each C–O bond in the resonance hybrid?

ANSWER: _____

19. Determine the hybridization of each atom (besides hydrogen) in CH_2O . Briefly justify each assignment using the local bonding environment of that atom.

ANSWER: _____

20. Use the number of unpaired electrons to classify NO as paramagnetic or diamagnetic, and state whether it is attracted to or repelled by a magnetic field. (NO has 1 unpaired electron in its molecular orbitals.)

ANSWER: _____

21. Rank the following nitrogen–nitrogen bonds from shortest to longest: N–N , N=N , $\text{N}\equiv\text{N}$. Explain your ranking.

ANSWER: _____

22. Which species is predicted to be more stable by MO theory: O_2 or O_2^{2-} ? Justify using bond orders.

ANSWER: _____

23. Rank the following ionic compounds in order of increasing lattice energy (smallest to largest). Briefly justify your ranking using Coulomb's law.

NaCl , LiCl , KCl

ANSWER: _____

24. Rank the following elements from highest metallicity to lowest metallicity. Support your ranking with an explanation of the underlying atomic structure; answers that simply cite a memorized periodic trend will not receive credit.

Ni, Fe, Cr, Cu

ANSWER: _____

25. Identify the hybridization of Cl in ClF_3 and describe the molecular geometry.

ANSWER: _____

26. Identify the main group to which X belongs, given the ionic compound Cr_2X_3 . Explain your answer.

ANSWER: _____

27. Consider the following compounds: phosphorus pentachloride, nitrogen dioxide, sulfur trioxide, and arsenic pentafluoride. Write the molecular formula for each of the listed compounds that has a dipole moment. If none have a dipole moment, write "None have a dipole moment". Draw each compound and explain your answer.

ANSWER: _____

28. Does C_2 exhibit s - p mixing in its molecular-orbital diagram? Explain.

ANSWER: _____

29. The species HCO_2^- exhibits resonance. Draw all of its equivalent resonance structures. What is the bond order of each C-O bond in the resonance hybrid?

ANSWER: _____

30. On the potential-energy curve for O_2 , what happens to the potential energy when the two atoms are very far apart? Why does this happen? Justify your answer.

ANSWER: _____

31. Using MO theory, determine the bond order of O_2 . Briefly justify your answer.

ANSWER: _____

32. Classify the bond between carbon (C) and sulfur (S) as nonpolar covalent, polar covalent, or ionic. If the bond is polar or ionic, identify which atom carries the partial negative charge (δ^-). Justify your answer using the electronegativity difference.

ANSWER: _____

33. Consider the species NH_4^+ . Draw its Lewis structure and use it to answer the following questions. How many total valence electrons must appear in the Lewis structure of NH_4^+ ? How many lone pairs surround the central nitrogen atom? Credit is awarded if you draw the Lewis structure correctly and answer both questions correctly.

ANSWER: _____

34. How many unhybridized p orbitals contribute to benzene's delocalized π system, and what do they form?

ANSWER: _____

35. Determine the total number of σ and π bonds in CH_3COCH_3 (acetone). For credit you must justify your answer by explaining the source of the σ and π bonds.

ANSWER: _____

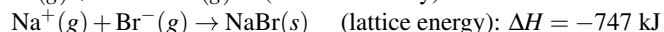
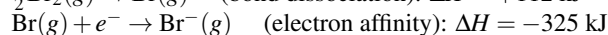
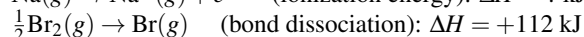
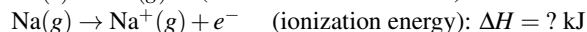
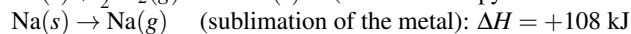
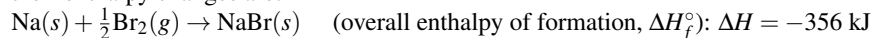
36. Draw the Lewis electron-dot symbol for an atom of Si.

ANSWER: _____

37. Identify the hybridization of Xe in XeF_4 and state the molecular geometry.

ANSWER: _____

38. Use a Born-Haber cycle (Hess's law) to determine the missing enthalpy change for the formation of NaBr(s) from its elements. The steps and their enthalpy changes are:

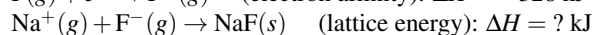
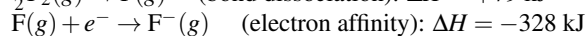
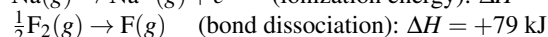
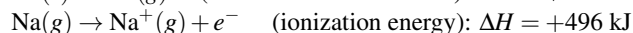
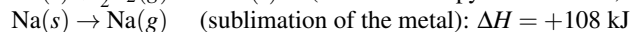


ANSWER: _____

39. Which bonding model is best suited for explaining why He₂ does not exist? Briefly justify your choice.

ANSWER: _____

40. Use a Born-Haber cycle (Hess's law) to determine the missing enthalpy change for the formation of NaF(s) from its elements. The steps and their enthalpy changes are:



ANSWER: _____

41. Rank the following bonds in order of increasing bond length: C – N, C ≡ N, C = N. Support your ranking with a brief explanation for credit. Credit is not awarded for simply citing a memorized trend.

ANSWER: _____

42. Identify the main group to which X belongs, given the ionic compound SrX₂. Explain your answer.

ANSWER: _____

43. A central atom in SiH_4 has 4 bonds and 0 lone pairs. Predict its hybridization, electron-group geometry, and molecular geometry. Explain how the number of electron groups and lone pairs determines each answer.

ANSWER: _____

44. Determine the electron-group arrangement, molecular shape, and ideal bond angle(s) for nitrogen dioxide.

ANSWER: _____

45. Consider the species ClO_3^- . Draw its Lewis structure and use it to answer the following questions. How many total valence electrons must appear in the Lewis structure of ClO_3^- ? How many lone pairs surround the central chlorine atom? Credit is awarded if you draw the Lewis structure correctly and answer both questions correctly.

ANSWER: _____

46. Determine the hybridization of C in CS_2 . State the molecular geometry and bond angle.

ANSWER: _____

47. Using MO theory, rank the following diatomic molecules in order of increasing bond order: N_2 , O_2 , F_2 . Justify your ranking by explaining how the MO electron occupancies lead to the observed trend.

ANSWER: _____

48. Sketch the valence molecular-orbital energy-level diagram for F_2 (14 valence electrons) and fill in its electrons. Determine the bond order and whether F_2 is paramagnetic or diamagnetic. Justify your conclusions.

ANSWER: _____

49. In valence bond theory, how does the geometry of orbital overlap determine the type of covalent bond that forms?

ANSWER: _____

50. MO theory predicts the same bond order for the following species: He_2^{2+} , H_2 . What is that bond order? Justify your answer using MO theory.

ANSWER: _____

51. Can two atoms be joined by a π bond without also sharing a σ bond? Explain by describing how σ and π bonds are formed. Explain your reasoning completely. Answers without a correct explanation will not receive full credit.

ANSWER: _____

52. A particular francium atom has 1 bond to another atom and has 0 lone pairs of electrons around it. Calculate the formal charge on this francium atom.

ANSWER: _____

53. Which bond is longer: the C–C single bond or the $\text{C}\equiv\text{C}$ triple bond? Explain.

ANSWER: _____

54. Classify the bond between lithium (Li) and fluorine (F) as nonpolar covalent, polar covalent, or ionic. If the bond is polar or ionic, identify which atom carries the partial negative charge (δ^-). Justify your answer using the electronegativity difference.

ANSWER: _____

55. Use molecular-orbital theory to determine the bond order and whether F_2^+ is paramagnetic or diamagnetic. Justify your answer.

ANSWER: _____

56. Is B_2^- paramagnetic or diamagnetic? Justify your conclusion using its molecular-orbital electron configuration.

ANSWER: _____

57. Rank the following bonds in order of increasing bond length: $\text{C}=\text{C}$, $\text{C}-\text{C}$, $\text{C}\equiv\text{C}$. Support your ranking with a brief explanation for credit. Credit is not awarded for simply citing a memorized trend.

ANSWER: _____

58. Identify the hybridization of S in SF_4 and describe the molecular geometry.

ANSWER: _____

59. In an atom, the valence $2s$ orbital lies at -24 eV and the $2p$ orbitals lie at -9 eV. When the atom is sp^2 -hybridized, each hybrid orbital has $\frac{1}{3}$ s -character and $\frac{2}{3}$ p -character. Treating the hybrid as a normalized linear combination, calculate the expectation value of its energy, $\langle E \rangle$.

ANSWER: _____

60. Draw the Lewis electron-dot symbol for an atom of Br.

ANSWER: _____

61. Determine the hybridization of each atom (besides hydrogen) in HCOOH . Briefly justify each assignment using the local bonding environment of that atom.

ANSWER: _____

62. Draw a valid Lewis structure for NO_2 . You must show all existing lone pairs and lone electrons.

ANSWER: _____

63. What is the effect on a molecule of placing electrons in (a) a bonding MO versus (b) an antibonding MO?

ANSWER: _____

64. Construct the valence molecular-orbital energy-level diagram for Be_2 (4 valence electrons) and fill in its electrons. Determine the bond order and whether Be_2 is paramagnetic or diamagnetic. Justify your conclusions.

ANSWER: _____

65. Which compound has a lower lattice energy, K_2S or K_2Se ? Explain your choice.

ANSWER: _____

66. Write the full electron configuration of a neutral aluminum atom. Then, using the octet rule, predict the charge of the monatomic ion that aluminum forms and identify the noble gas with which that ion is isoelectronic. Explain your reasoning.

ANSWER: _____

67. A diatomic molecule has a measured dipole moment of 2.21 D and a bond length of 107 pm. Calculate the percent ionic character of the bond. Report your answer with two digits after the decimal.

ANSWER: _____

68. An atom forms three equivalent bonds arranged 120° apart. Is this explained primarily by hybridization or by σ/π bonding? Justify your answer.

ANSWER: _____

69. Rank H_2^- , H_2^+ , and H_2 from least to most strongly attracted to a magnetic field. Justify your ranking by writing the molecular-orbital electron configuration of each species.

ANSWER: _____

70. Write the complete MO electron configuration for O_2 . Determine its bond order and magnetic character.

ANSWER: _____

71. Determine the hybridization of C in CH_4 , given that it has 4 bonds and 0 lone pairs of electrons. What is the molecular geometry?

ANSWER: _____

72. Rank the following elements from highest metallicity to lowest metallicity. Support your ranking with an explanation of the underlying atomic structure; answers that simply cite a memorized periodic trend will not receive credit.

Ga, Ti, Ge, As

ANSWER: _____

73. Which species is predicted to be more stable by MO theory: N_2 or O_2 ? Justify using bond orders.

ANSWER: _____

74. Use average bond energies to estimate ΔH° for the reaction $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, and state whether the reaction is endothermic or exothermic. Average bond energies (kJ/mol): $\text{H}-\text{H} = 436$, $\text{N}\equiv\text{N} = 945$, $\text{N}-\text{H} = 391$.

ANSWER: _____

75. When the $2p$ orbitals of two atoms combine, how many antibonding molecular orbitals are formed? Justify your answer.

ANSWER: _____

76. Which atom has the larger atomic radius: caesium (Cs) or lithium (Li)? You must briefly explain why. Credit is awarded only for explaining the cause of the trend based on atomic structure, not for merely stating the trend.

ANSWER: _____

77. Using MO theory, rank the following diatomic molecules in order of increasing bond order: C_2 , N_2 , F_2 . Justify your ranking by explaining how the MO electron occupancies lead to the observed trend.

ANSWER: _____

78. Using MO theory, determine the bond order of Li_2^+ . Briefly justify your answer.

ANSWER: _____

79. In orbital drawings, what do the $+$ and $-$ signs (or the two colors) on a wavefunction represent — and what do they NOT represent?

ANSWER: _____

80. Write the complete MO electron configuration for B_2 . Determine its bond order and magnetic character.

ANSWER: _____

81. Does the central atom in CCl_4 obey the octet rule? If not, classify the exception as electron-deficient or expanded octet, and state the number of electrons around the central atom.

ANSWER: _____

82. Rank the following ionic compounds in order of increasing lattice energy (smallest to largest). Briefly justify your ranking using Coulomb's law.
KF, LiF, NaF

ANSWER: _____

83. A diatomic molecule has a measured dipole moment of 2.27 D and a bond length of 128 pm. Calculate the percent ionic character of the bond. Report your answer with two digits after the decimal.

ANSWER: _____

84. Determine the hybridization of C in CH_2O . State the molecular geometry and approximate bond angle.

ANSWER: _____

85. True or false: We apply the hybridization model to isolated, unbonded atoms. Explain.

ANSWER: _____

86. Which bonding model is best suited for describing electrons delocalized over an entire molecule? Briefly justify your choice.

ANSWER: _____

87. Identify the hybridization of each carbon atom in $CH_3 - C(=O) - CH_3$ (acetone). Explain your answer.

ANSWER: _____

88. Compare H_2^+ and H_2 using molecular-orbital (MO) theory. For each species, write the MO electron configuration, determine the bond order and magnetic character, and then explain how removing one electron affects the bond order and the magnetic behavior.

ANSWER: _____

89. Use molecular-orbital theory to determine the bond order and whether B_2^+ is paramagnetic or diamagnetic. Justify your answer.

ANSWER: _____

90. Experimental measurements show that the bond angle(s) in SO_3 are 120° . Explain why this observation led chemists to describe the bonding on the S atom using hybrid orbitals.

ANSWER: _____

91. Resonance structures for SO_3 differ from one another. Explain what actually changes between resonance structures and why those changes do not alter the hybridization of the central atom. Explain your reasoning completely. Your explanation must identify the feature(s) that determine hybridization and why resonance does or does not affect them. Answers without this explanation will not receive full credit.

ANSWER: _____

92. Compare Li_2^+ and Li_2 using molecular-orbital (MO) theory. For each species, write the MO electron configuration, determine the bond order and magnetic character, and then explain how removing one electron affects the bond order and the magnetic behavior.

ANSWER: _____

93. What real-world consequence does the restricted rotation about $\text{C}=\text{C}$ double bonds have for the behavior of unsaturated vs. saturated fats?

ANSWER: _____

94. Between the same two atoms, which is the stronger bond — a σ bond or a π bond? Explain.

ANSWER: _____

95. Why do O_2 , F_2 , and Ne_2 NOT show s - p mixing, while Li_2 through N_2 do?

ANSWER: _____

96. A molecule's central atom is surrounded by 6 bonding groups and 0 lone pairs of electrons. Determine the electron-group arrangement, the molecular shape, and the ideal bond angle(s).

ANSWER: _____

97. A molecule's central atom is surrounded by 6 bonding groups and 0 lone pairs of electrons. Determine the electron-group arrangement, the molecular shape, and the ideal bond angle(s).

ANSWER: _____

98. On the potential-energy curve for O₂, what happens to the potential energy when the two atoms are very far apart? Why does this happen? Justify your answer.

ANSWER: _____

99. Two atoms form the same number of bonds but differ in whether one bond can be rotated without changing the bonding interaction. Is this difference explained primarily by hybridization or by σ/π bonding? Justify your answer.

ANSWER: _____

100. Determine the electron-group arrangement, molecular shape, and ideal bond angle(s) for xenon tetrafluoride.

ANSWER: _____